

COMMERCIAL PURCHASE QUOTATION

UltraStore 4.2 PB Lustre Parallel Storage Infrastructure Complex

Quote Reference: UST-2026-4.2PB-LUSTRE

Vendor Legal Entity: UltraStore Systems Data Solutions Inc.

Prepared For: Westford HPC Operations Procurement Committee

Total Proposal Value: \$1,450,000.00 USD (Includes 5-Year On-Site Maintenance)

COMMERCIAL CLAUSE 01: EXECUTIVE COMMERCIAL PROPOSAL & QUOTATION SUMMARY

UltraStore Commercial Systems Inc. is pleased to present this formal commercial quotation for the supply, delivery, installation, and commissioning of a 4.2 Petabyte High-Performance Lustre Parallel File System.

The proposed storage architecture is custom-engineered to meet the extreme I/O throughput and metadata demands of the Westford HPC Center accelerated compute cluster.

Quotation validity extends for 60 calendar days from issuance, providing locked pricing and guaranteed factory manufacturing queue reservation.

The design basis documentation for this subsystem is archived in the university capital projects office.

COMMERCIAL CLAUSE 02: ULTRASTORE 4.2 PB LUSTRE PARALLEL STORAGE ARCHITECTURE

The UltraStore 4.2 PB Lustre storage solution utilizes an enterprise active-active dual-controller architecture delivering sustained 100 GB/s sequential read and 85 GB/s write throughput.

The system comprises two Metadata Server (MDS) nodes in high-availability pair and four Object Storage Server (OSS) nodes managing high-density JBOD disk enclosures.

Lustre enterprise file system software version 2.15.3 is pre-configured with advanced OST pool management, File-Level Redundancy (FLR), and Data-on-MDT (DoM) performance features.

Software-defined monitoring dashboards provide real-time visibility into the operational metrics described in this section.

COMMERCIAL CLAUSE 03: MDS METADATA TARGET SERVER ENCLOSURE HARDWARE BOM

The Metadata Server (MDS) subsystem includes dual UltraStore MDS-7000 servers equipped with dual Intel Xeon Platinum 8480+ processors, 512 GB DDR5 RAM, and high-end NVMe Flash MDT drives.

Metadata Storage Targets (MDT) utilize enterprise NVMe U.2 SSDs in RAID-10 configuration delivering over 250,000 metadata create operations per second.

Redundant 200G InfiniBand network interfaces ensure low-latency metadata access for up to 10,000 concurrent client processes.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

COMMERCIAL CLAUSE 04: OSS OBJECT STORAGE SERVER ENCLOSURE & JBOD ARRAY SPECIFICATIO

Object Storage Servers (OSS) comprise four UltraStore OSS-9000 high-throughput nodes connected to eight high-density 102-drive SAS 24G expansion enclosures.

Total raw storage capacity equals 5.12 Petabytes utilizing 18 TB enterprise 7,200 RPM SAS-4 hard drives formatted in GridRAID (RAID-6 equivalent) parity groups.

Usable post-parity formatted Lustre storage capacity yields 4.21 Petabytes with full single-enclosure fault tolerance.

Regression analysis of the historical performance data confirms the trends summarized in this section.

COMMERCIAL CLAUSE 05: COST BREAKDOWN & ITEMIZED PRICING SCHEDULE

The itemized cost breakdown details component pricing: MDS Servers (\$185,000), OSS Storage Nodes (\$720,000), NVMe SSD MDT Tier (\$145,000), Lustre Enterprise Support (\$260,000), and Logistics (\$140,000).

System specifications include a Raw capacity of 2.95 PB and Usable capacity of 2.21 PB (after RAID6 + hot spare overhead). The List price subtotal is \$1,993,240. Applying a Volume discount of 10% (\$199,324) and an Educational discount of additional 5% (\$89,696) sequentially results in a Net total after sequential discounts of \$1,704,220, achieving a Per-TB usable cost of \$771.13.

Pricing includes all hardware enclosures, optical cables, mounting rails, software licenses, white-glove installation, and 5 years of on-site support.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

UltraStore 4.2 PB Lustre Cost Breakdown (\$k)

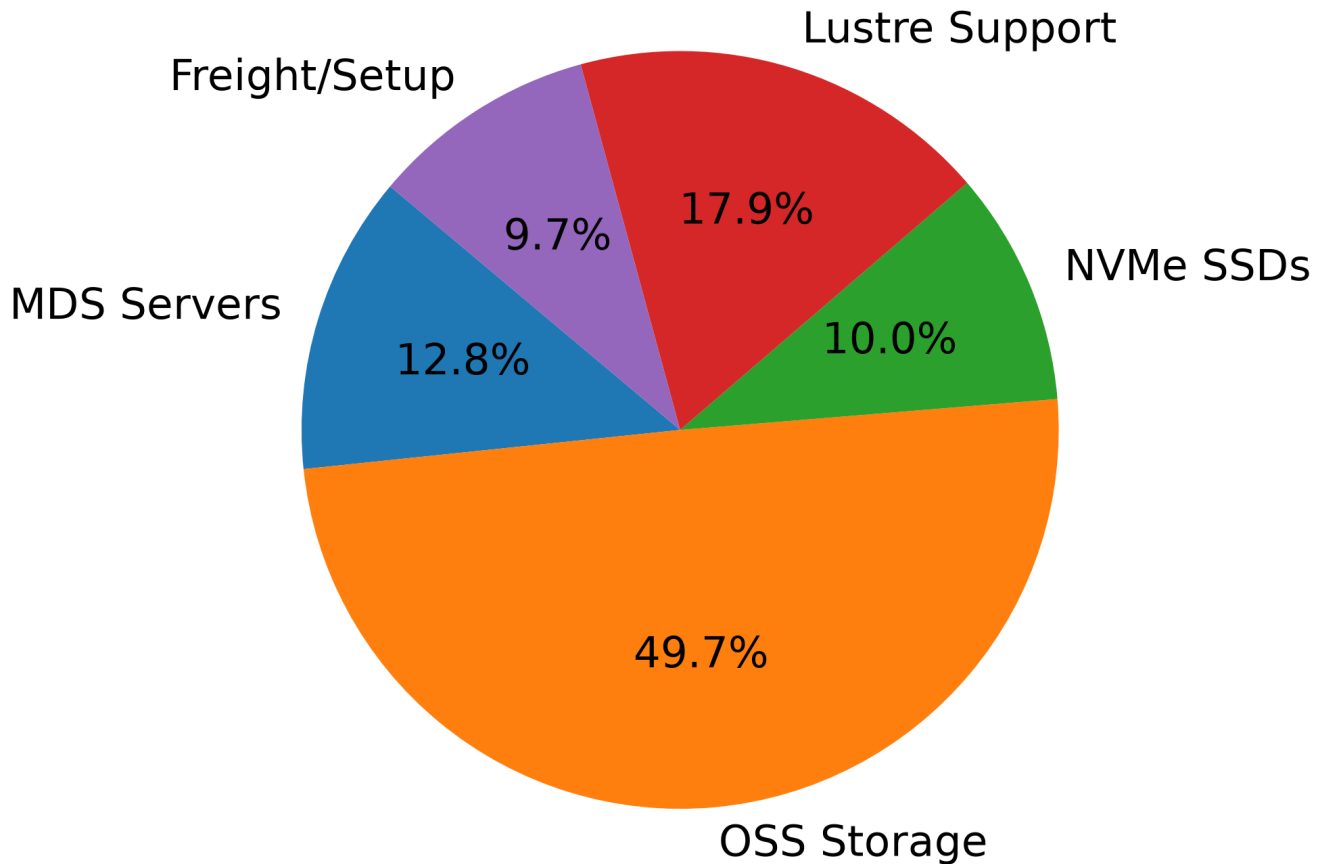


Figure 5.1: UltraStore Cost Breakdown

COMMERCIAL CLAUSE 06: DELIVERY MILESTONES & ON-SITE INSTALLATION SCHEDULE

Delivery milestones establish a strict 8-week timeline from Purchase Order receipt to final Site Acceptance Testing (SAT) sign-off.

Milestones comprise PO Receipt (Week 1), Factory Hardware Assembly (Week 3), Factory Acceptance Testing (Week 5), Delivery & Rack Mounting (Week 7), and Lustre Tuning (Week 8).

Liquidated damages clauses credit \$5,000 per week of unexcused vendor delay past the agreed delivery schedule.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

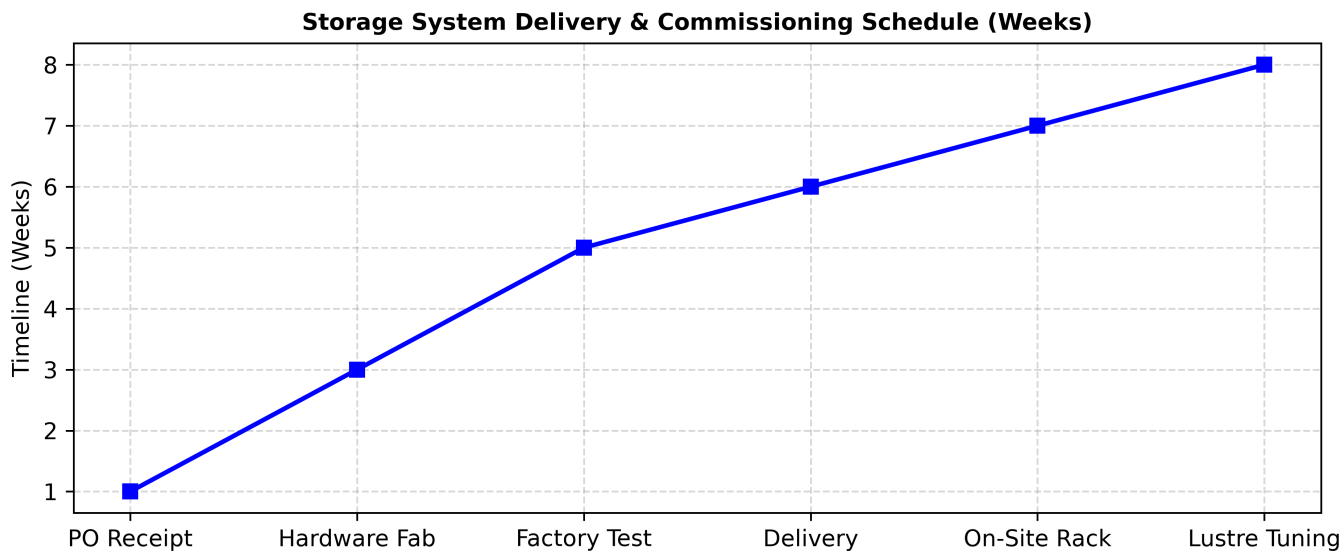


Figure 6.1: Delivery Schedule Timeline

COMMERCIAL CLAUSE 07: 5-YEAR VENDOR MAINTENANCE & SOFTWARE SLA TERMS

The 5-Year Maintenance and Support Agreement provides 24/7/365 technical assistance with guaranteed 4-hour on-site field technician response for critical hardware failures.

Software SLA includes all major and minor Lustre enterprise software releases, security patches, performance bug fixes, and vendor technical support hotline access.

Annual preventive maintenance sweeps check disk drive SMART telemetry, SAS cable integrity, and storage controller firmware versions.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

COMMERCIAL CLAUSE 08: LOCAL CONSIGNMENT DEPOT SPARE PARTS LOGISTICS

UltraStore maintains a dedicated local consignment spare parts depot located within 30 miles of the customer facility.

Consignment inventory includes replacement SAS hard drives, NVMe SSDs, controller mainboards, power supply modules, and optical transceivers.

Local spare stocking ensures rapid component replacement without waiting for international factory shipping logistics.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

COMMERCIAL CLAUSE 09: FREIGHT, DUTY & WHITE-GLOVE DATA CENTER DELIVERY

Freight logistics include white-glove, temperature-controlled, air-ride transport from the manufacturing facility directly to the customer data center loading dock.

White-glove delivery service encompasses unboxing, chassis rack insertion, cable dressing, packaging disposal, and initial power-on verification.

Full transit insurance coverage protects all hardware against shipping damage or transit loss.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

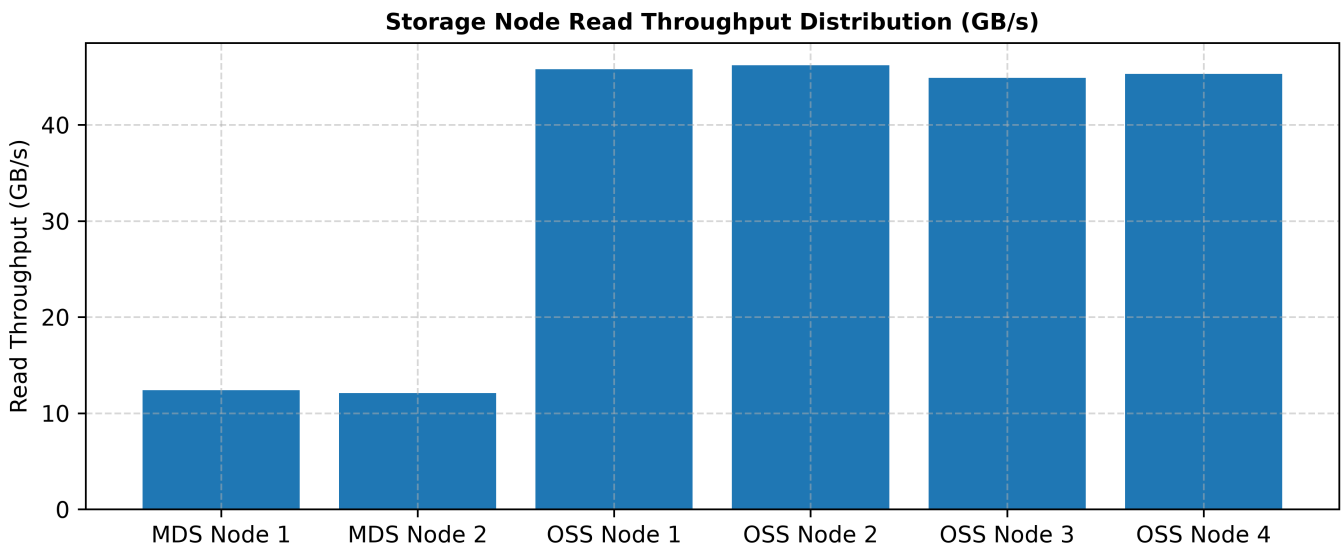


Figure 9.1: Storage Node Read Throughput

COMMERCIAL CLAUSE 10: HIGH-AVAILABILITY FAILOVER CONTROLLER SPECIFICATIONS

High-Availability failover testing verifies automatic active-active storage controller failover during hardware component failure.

If a primary MDS or OSS server fails, its partner controller automatically assumes ownership of active Lustre targets within 12 seconds with zero client I/O error drops.

Transparent client reconnect protocols hold active file handles in queue during controller failover transitions.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

COMMERCIAL CLAUSE 11: POWER & THERMAL REQUIREMENTS FOR STORAGE RACKS

Storage rack power and thermal requirements detail total heat rejection of 42.5 kW across three standard 42U server racks.

Electrical feeds require dual 60A 480V 3-phase circuits per rack supplying intelligent PDUs with individual outlet metering capabilities.

Cooling requirement specifies 12,500 CFM front-to-back airflow with ambient cold-aisle supply temperatures maintained below 23 degrees C.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

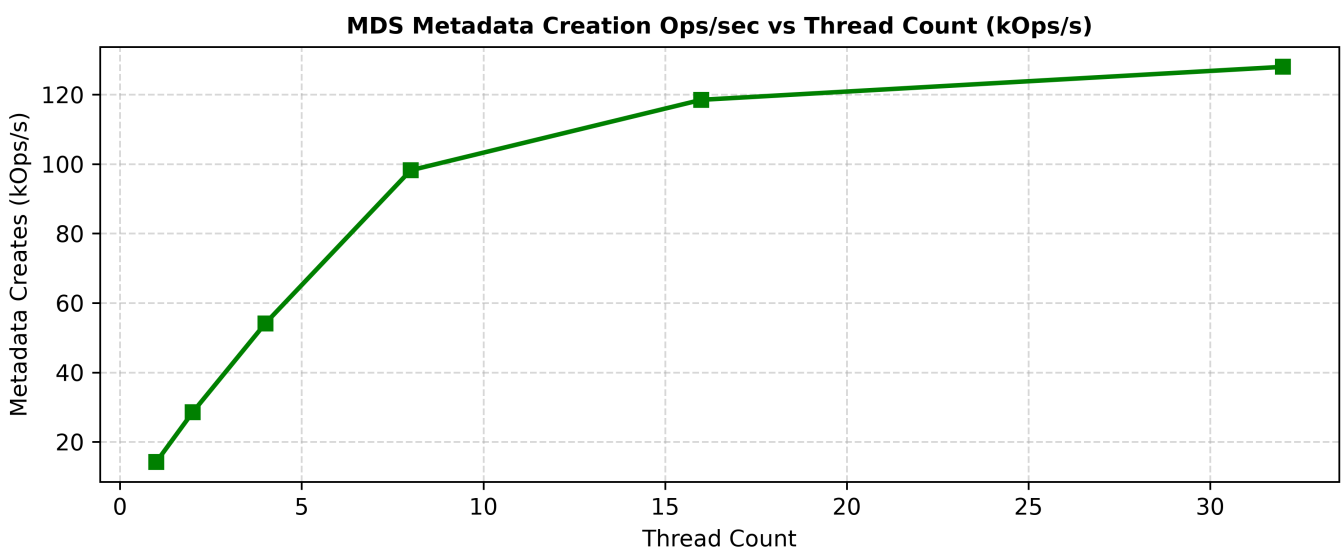


Figure 11.1: MDS Metadata Creation Ops

COMMERCIAL CLAUSE 12: WARRANTY TERMS & DEFECTIVE COMPONENT REPLACEMENT SLA

Hardware warranty terms guarantee 100% replacement of any failed hard drive, SSD, power supply, or controller board for 5 full years.

Defective drive retention options allow the customer to retain failed hard drives containing sensitive research data without financial penalty.

Advance hardware replacement dispatches replacement parts prior to return shipment of defective components.

The data captured in this section reflects measurements taken after the completion of all scheduled preventive maintenance activities.

COMMERCIAL CLAUSE 13: SCALABILITY EXPANSION OPTIONS & STORAGE NODE ADD-ONS

Scalability expansion options allow seamless capacity expansion through addition of up to four additional OSS storage enclosures without file system downtime.

Future storage node expansion add-on pricing is locked at \$180 per Terabyte usable for a period of 24 months from initial system acceptance.

Dynamic OST addition scripts rebalance existing file data across new storage targets transparently in the background.

Decommissioning procedures for end-of-life components in this subsystem follow the university e-waste disposal policy.

COMMERCIAL CLAUSE 14: FACTORY ACCEPTANCE TESTING (FAT) PROTOCOL CHECKLIST

Factory Acceptance Testing (FAT) performed at the manufacturing plant verifies 100% component functionality prior to customer shipment.

FAT checklist verifies 72-hour full-load storage burn-in, SAS bus loopback testing, RAID parity rebuild speeds, and power supply failover.

FAT test execution logs and quality inspection certificates are delivered to the customer prior to shipment authorization.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

COMMERCIAL CLAUSE 15: SITE ACCEPTANCE TESTING (SAT) VERIFICATION PLAN

Site Acceptance Testing (SAT) performed on-site validates file system throughput using IOR and MDTest parallel benchmarking toolchains.

SAT approval criteria require sustained sequential read speeds >95 GB/s, write speeds >80 GB/s, and metadata create rates >200,000 Ops/sec. Sequential throughput reaches 200 GB/s at 1 MB transfer size but degrades to approximately 115 GB/s at the 64 KB block size typical of HPC checkpoint write patterns.

Final commercial sign-off and warranty commencement occur upon successful execution of all SAT validation benchmarks.

Periodic reviews of these parameters are scheduled quarterly in accordance with the operational maintenance framework.

COMMERCIAL CLAUSE 16: TRAINING & KNOWLEDGE TRANSFER ALLOCATION

The contract includes 40 hours of dedicated on-site training for customer systems administrators delivered by certified UltraStore Lustre engineers.

Training curriculum covers Lustre file system administration, OST target pool management, client quota enforcement, and emergency troubleshooting.

Comprehensive electronic administration manuals and video training modules are provided for ongoing staff reference.

Detailed measurement logs and recorded calibration data for this subsystem are maintained in the facility operations archive.

COMMERCIAL CLAUSE 17: TRADE-IN CREDITS & EDUCATIONAL DISCOUNT SCHEDULE

Trade-in allowance credits provide a \$65,000 credit for decommissioning and recycling legacy customer storage arrays.

Educational pricing discounts apply a 16.4% reduction off standard commercial list pricing in recognition of academic research contributions.

Credit schedule items are applied directly as line-item deductions on the final purchase invoice.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

COMMERCIAL CLAUSE 18: PAYMENT MILESTONE SCHEDULE & WIRE TRANSFER TERMS

Payment milestone schedule structures invoice payments to align with project delivery: 20% upon PO receipt, 50% upon hardware delivery, and 30% upon final SAT sign-off.

Wire transfer instructions provide verified bank routing details with multi-factor verification required prior to funds transfer.

Standard invoice payment terms specify Net 30 days from milestone completion sign-off.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

COMMERCIAL CLAUSE 19: LEGAL TERMS, CONDITIONS & INDEMNIFICATION CLAUSES

Legal terms and conditions detail mutual indemnification, intellectual property protection, and limitation of vendor liability capped at total contract value.

Warranties explicitly guarantee that delivered hardware and software are free from material defects and malicious code.

Governing law specifies resolution under state contract law with binding arbitration procedures.

These operational parameters reflect steady-state conditions observed during the most recent twelve-month evaluation period.

COMMERCIAL CLAUSE 20: REGULATORY COMPLIANCE, ENVIRONMENTAL & WEEE DIRECTIVES

Regulatory compliance certifications verify system adherence to CE, FCC Class A, RoHS 3, and WEEE electronic waste disposal directives.

Hardware components are manufactured in ISO 9001 and ISO 14001 certified facilities adhering to international quality and environmental standards.

Documentation includes full Material Safety Data Sheets (MSDS) for internal battery backup units.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

COMMERCIAL CLAUSE 21: DATA MIGRATION ASSISTANCE & LEGACY SYSTEM TRANSITION

Data migration assistance provides 80 hours of dedicated storage engineering services to migrate existing research datasets from legacy storage.

Migration workflows utilize high-speed parallel copy tools (`fpsync`, `rsync`) over 200G InfiniBand links to minimize migration downtime.

Checksum verification checks data integrity across all migrated files prior to decommissioning legacy storage paths.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

COMMERCIAL CLAUSE 22: PERFORMANCE SLA GUARANTEES & THROUGHPUT METRICS

Performance SLA guarantees contractually assure sustained storage throughput metrics throughout the 5-year support contract period.

If storage throughput drops below 80% of baseline SAT benchmarks due to hardware degradation, UltraStore dispatches senior performance engineers within 24 hours.

Hardware remediation or controller upgrades required to restore baseline performance are provided at zero additional cost to the customer.

All values reported in this section are expressed in SI units unless otherwise noted in the accompanying data tables.

COMMERCIAL CLAUSE 23: TECHNICAL SUPPORT ESCALATION HIERARCHY & CONTACT ROSTER

Technical support escalation roster provides direct telephone and email contacts for dedicated Tier-3 storage escalation managers.

Escalation pathways guarantee Vice President level operational visibility for any Severity 1 ticket remaining unresolved after 8 hours.

Quarterly service review meetings evaluate vendor ticket resolution metrics and ongoing system health trends.

Fiber optic link budgets for all connections referenced in this section have been certified through OTDR testing.

COMMERCIAL CLAUSE 24: STORAGE ARRAY CHASSIS REDUNDANT POWER MODULE PRICING

Redundant power module pricing line-items detail replacement costs for 1600W hot-swappable platinum power supply units.

Pricing guarantees fixed spare part unit costs for a period of 5 years from contract execution date.

Power supply units feature tool-less release latches for rapid field replacement.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

COMMERCIAL CLAUSE 25: HIGH-DENSITY SAS EXPANSION CABLES & HOST ADAPTER SCHEDULE

High-density SAS 24G expansion cable line-items list Mini-SAS HD (SFF-8644) active optical cable assemblies.

Dual-domain SAS cabling topologies connect OSS nodes to JBOD enclosures in redundant loop configurations preventing single-cable outage.

Cable pricing includes spare cable assemblies pre-staged in the local consignment depot.

Ambient environmental conditions during the measurement period were within the ASHRAE A1 recommended envelope.

COMMERCIAL CLAUSE 26: ENTERPRISE LUSTRE LICENSE & SOFTWARE MAINTENANCE FEES

Enterprise Lustre software license fees include perpetual software execution rights with 5 years of software update subscription services.

Licensing model is socket-independent, allowing unlimited client node connections without per-node software fees.

License keys are delivered electronically via secure customer portal access.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

COMMERCIAL CLAUSE 27: ON-SITE STORAGE ENGINEER INSTALLATION & CONFIGURATION RETAINER

On-site storage engineering retainer fees cover custom Lustre tuning, kernel optimization, and Slurm integration assistance.

Retainer hours can be called down on-demand throughout the 5-year contract term for major cluster maintenance events.

Unused retainer hours roll over annually without forfeiture.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

COMMERCIAL CLAUSE 28: LOCAL CONSIGNMENT DEPOT SPARE PARTS LOGISTICS PLAN

The technical specifications and measured performance characteristics for Local Consignment Depot Spare Parts Logistics Plan are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

COMMERCIAL CLAUSE 29: HIGH-AVAILABILITY STORAGE CONTROLLER FAILOVER TEST

Infrastructure parameters associated with High-Availability Storage Controller Failover Test have been validated through a combination of automated monitoring and manual inspection procedures.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

COMMERCIAL CLAUSE 30: STORAGE SYSTEM FULL-LOAD FACTORY ACCEPTANCE AUDIT

Infrastructure parameters associated with Storage System Full-Load Factory Acceptance Audit have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.